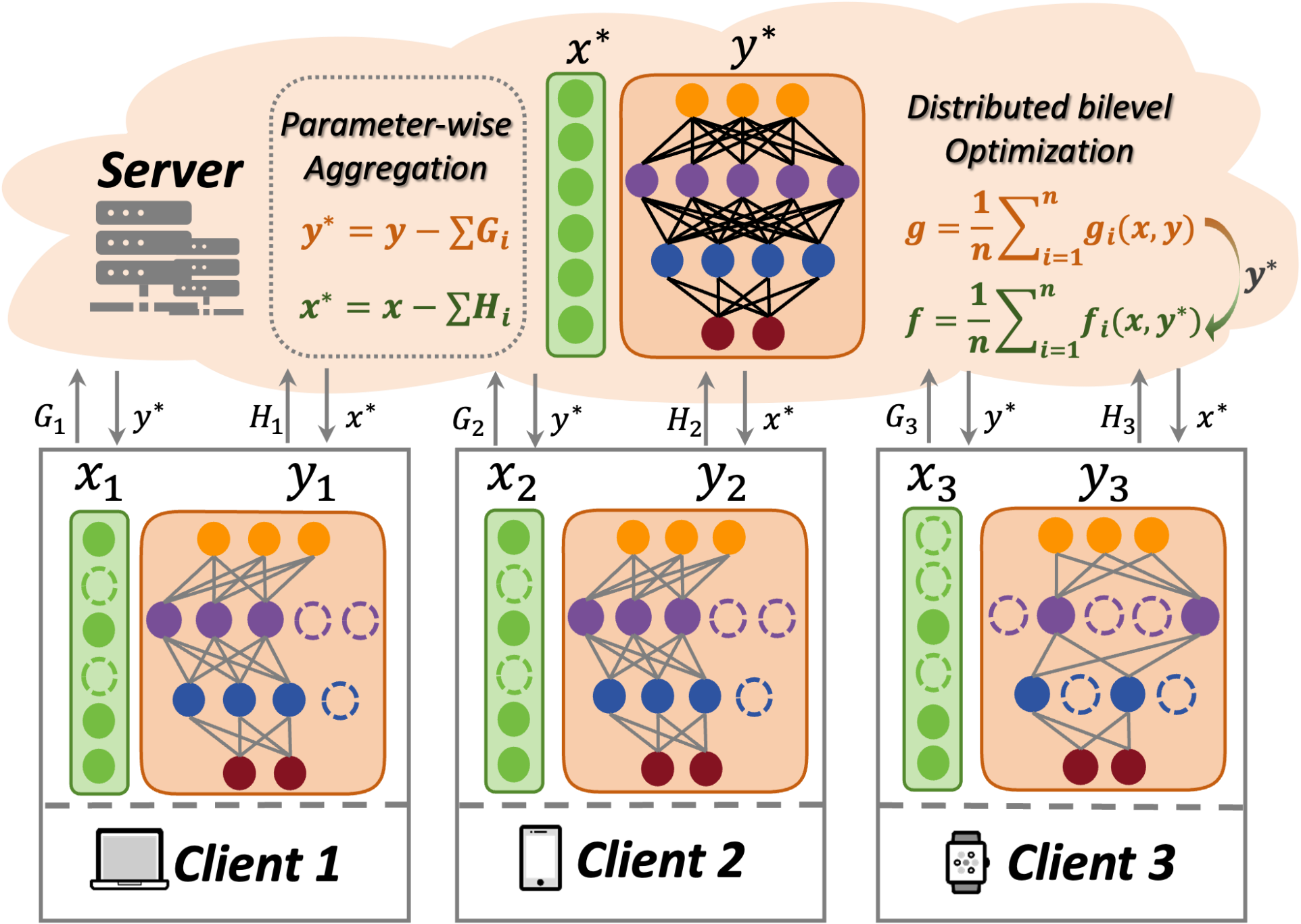
Server
Parameter-wise Aggregation
$y^* = y - \sum G_i$
$x^* = x - \sum H_i$
x^*
y^*
Distributed bilevel Optimization
$g = \frac{1}{n}\sum_{i=1}^{n} g_i(x, y)$
$f = \frac{1}{n}\sum_{i=1}^{n} f_i(x, y^*)$
y^*
G_1
y^*
H_1
x^*
G_2
y^*
H_2
x^*
G_3
y^*
H_3
x^*
x_1
y_1
x_2
y_2
x_3
y_3
Client 1
Client 2
Client 3